

Screenshots for how a ui is created and its properties

- Created through external embedded html form

The screenshot shows the Camunda BPMN editor interface. On the left, a list of files includes 'Assignment3.bpmn', 'billForm.html', 'draftReviewForm.html', 'draftUploadForm.html' (highlighted), 'materialListForm.html', and 'requirementsForm.html'. On the right, the 'draftUploadForm.html' form is displayed. It features a 'Requirements History' section with a list of items, each containing a date and requirements. Below this is an 'Upload the draft' section with a file upload button. The 'Raw Data' section shows the underlying HTML and JavaScript code for the form, including Camunda-specific attributes like 'cam-script', 'cam-form', and 'cam-variable-name'.

```
draftUploadForm.html
{{ history.length ? "New requirements" : "Requirements" }} from user {{userName}}
{{ current.requirements }}

Requirements History
  • {{ x.date.toString() }}: {{ x.requirements }}

Upload the draft
 未选择任何文件

Raw Data:
<form role="form" name="form">
  <script cam-script type="text/form-script">
    camForm.on('form-loaded', function() {
      camForm.variableManager.fetchVariable('history');
      camForm.variableManager.fetchVariable('userName');
    });
    camForm.on('variables-fetched', function() {
      $scope.history = JSON.parse(camForm.variableManager.variableValue('history'));
      $scope.userName = camForm.variableManager.variableValue('userName');
      $scope.current = $scope.history.shift();
    });
  </script>
  <b>{{ history.length ? "New requirements" : "Requirements" }} from user {{userName}}</b>
  <p>{{ current.requirements }}</p>
  <div ng-if="history.length">
    <label>Requirements History</label>
    <ul>
      <li ng-repeat="x in history">
        {{ x.date.toString() }}: {{ x.requirements }}
      </li>
    </ul>
    <a cam-file-download="draft"></a>
  </div>
  <div class="form-group">
    <label for="draft-field">Upload the draft</label>
    <input type="file" cam-variable-name="latestDraft" cam-variable-type="File" cam-max-file-size="10000000" class="form-control" />
  </div>
</form>
```

- Code example: bind properties through cam-variable-name attribute or JS & Angular code

```
<form role="form" name="form">
  <script cam-script type="text/form-script">
    camForm.on('form-loaded', function() {
      camForm.variableManager.fetchVariable('history');
    });
    camForm.on('variables-fetched', function() {
      $scope.history = JSON.parse(camForm.variableManager.variableValue('history'));
    });
  </script>
  <div class="form-group">
    <label for="requirements-field">{{ history.length ? "New requirements" : "Requirements" }}</label>
    <textarea required cam-variable-name="newRequirements" cam-variable-type="String">
  </div>
  <div ng-show="history.length">
    <label>Requirements History</label>
    <ul>
      <li ng-repeat="x in history">
        {{ x.date.toString() }}: {{ x.requirements }}
      </li>
    </ul>
  </div>
</form>
```

Screenshots for how variables are passed

- Inline script to declare variables when necessary

 **USER TASK**
Collect requirements

Execution listeners + 1 ▼

▼ End: Script

Event type

end ▼

Listener type

Script ▼

Format

JS

Type

Inline script ▼

Script

```
const requirements = execution.getVariable("newRequirements");
const date = new Date();
const history = JSON.parse(execution.getVariable("history"));
history.unshift({ date, requirements });
execution.setVariable("history", JSON.stringify(history));
execution.setVariable("newRequirements", "");
```

Field injection +

- JS external script for service tasks

```
client.subscribe("generateMaterialList", async function({ task, taskService }) {
  const n = Math.floor(Math.random() * 4 + 2);
  const materialList = new Array();
  for (var i = 0; i < n; i++) {
    materialList.push("Material" + i);
  }
  processVariables.set("materialList", JSON.stringify(materialList));
  await taskService.complete(task, processVariables);
});

client.subscribe("generateBill", async function({ task, taskService }) {
  const refineTimes = task.variables.get("refineTimes");
  const buildMethod = task.variables.get("buildMethod");
  const designFee = (350 + 50 * refineTimes);
  const buildFee = buildMethod === 'company' ? 3000 : 0;
  processVariables.set("bill", `Design fee: ${designFee}€\nBuild Fee: ${buildFee}€`);
  await taskService.complete(task, processVariables);
});
```

Screenshots for each user task

● Collect requirements

For the first time

Form

History

Diagram

Description

Name

ga94hor

Requirements

R1, R2, R3

Save

Complete

Add additional requirements

Name

ga94hor

New requirements

R5

Requirements History

- 2022-06-19T13:23:25.967Z: R4
- 2022-06-19T13:19:18.576Z: R1, R2, R3

Save

Complete

● Upload the draft

Form

History

Diagram

Description

New requirements from user ga94hor

2022-06-19T13:26:01.469Z: R5

Requirements History

- 2022-06-19T13:23:25.967Z: R4
- 2022-06-19T13:19:18.576Z: R1, R2, R3

Upload the draft

选择文件

 draft.pdf

Save

Complete

● Review the draft

if not satisfied with the draft

Form	History	Diagram	Description
Review the draft draft.pdf Are you satisfied with the draft? ☐ Yes ☒ No, I want to continue to refine the draft. Save Complete			

if satisfied with the draft, then make decision about the build method

Review the draft draft.pdf Are you satisfied with the draft? ☒ Yes ☐ No, I want to continue to refine the draft. How would you like to build the house? ☐ Entrust it to the company ☒ Build it yourself. Save Complete
--

● Buy materials

if the user want to build the house by himself

Form	History	Diagram	Description
Material List - Material0 - Material1 - Material2 - Material3 - Material4 Have all the needed materials been bought? ☐ Yes ☒ No Save Complete			

● Pay the bill

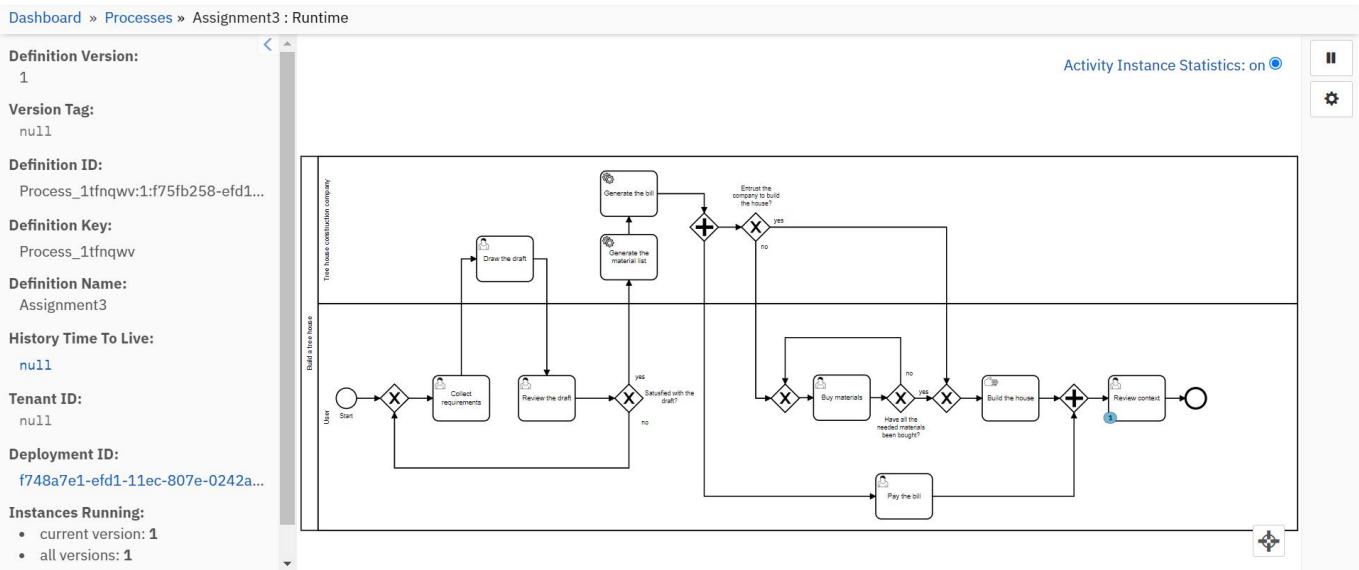
FormHistoryDiagramDescription

Please pay your bill

Design fee: 450€
Build Fee: 0€
Total: 450€

SaveComplete

Screenshot of the running process



Screenshot of the last task displaying the context

Add a variable +			
	Name	Type	Value
	refineTimes	Integer	2
	satisfied	Boolean	☒
	boughtAll	Boolean	☒
Remove x	draft	File	Download
	materialList	String	["Material0","Material1","Material2","Material3","M:
	buildMethod	String	client
	bill	String	Design fee: 450€Build Fee: 0€Total: 450€
	history	String	[{"date":"2022-06-19T13:26:01.469Z","requireme
	userName	String	ga94hor
	newRequirements	String	Value
			Complete